**7.** $f(x) = Ax^3 + 6x^2 - 4x + B$

where $A$ and $B$ are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$
- $\int_3^5 f(x)\,dx = 176$

find the value of $A$ and the value of $B$.

**(7)**